

Image Compression Using Hybrid Radon Transform with Discrete Wavelet Transform Technique



R. Nanmaran, K. Vishalatchi, S. Srimathi, and V. Vinoth Kumar

Abstract Image compression is the method of decreasing the image measure without influence the quality of input image. A hybrid transform-based image compression has been proposed in this work. In this work, a Radon transforms (RT) with discrete wavelet transform (DWT) has been used for image compression technique. Performance of proposed algorithm is measured using high values of Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Compression Ratio (CR). The proposed method provides PSNR of 28.73 dB, SSIM of 89.41%, and CR of 88.33%. The proposed cross breed procedure (RT + DWT) gives way better compression comes about when compared to compression calculations utilized exclusively which are apparent from the reenactment comes about. When compared to existing state of crossover method gives competing comes about and this strategy can be utilized for areas like medication, obsequious communication, huge information analytics, and other areas. Where requires more information but with less memory space.

Keywords Radon transform (RT) · Discrete wavelet transform (DWT) · Compression Ratio (CR) · Peak signal-to-noise ratio (PSNR) · Structural similarity index measure (SSIM)

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1 Introduction

Image compression is the method of lessening the estimate of advanced pictures without degrading their quality and basic data information. One commonly utilized strategy of picture compression is the utilization of the Radon transform in combination with the discrete wavelet transform [1]. The Radon transform is a mathematically technique that represents an image as a collection of line integrals. It is commonly used in medical imaging and computer vision applications. The Radon transform can be utilized to extricate highlights from a picture and to compress it by speaking to it in a more compact shape [2]. The discrete wavelet transform (DWT) is another broadly utilized method for image compression. It breaks down an image into a set of wavelet coefficients at diverse scales and introductions. The wavelet coefficients can be utilized to represent the image in a compressed form while retaining its essential information. By combining the Radon transform and the DWT, can obtain a more efficient image compression algorithm [3]. The Radon transform is utilized to extricate highlights frame the picture, and the DWT is utilized to speak to these highlights in a compressed shape. The coming about compressed picture can be reproduced by applying the inverse DWT and Radon transform. This approach has been shown to be effective in compressing medical images, such as CT scans and MRI images, as well as natural images [4]. It can to be utilized in other applications, such as video compression and information capacity.

2 Literature Review

Pal et al. [5] proposed a image compression technique using radon transform for medical images. The proposed method is also evaluated in terms of computational complexity to assess its suitability for real-time image compression application. In the background of this study is the need for efficient image compression techniques and the potential benefits of combining the Radon transform with the DWT [6]. The study aims to investigate the performance of proposed algorithm and evaluate in terms of Compression Ratio, SSIM, and PSNR. Image compression is an important aspect of image processing that involves reducing the size of an image while maintaining its quality [7]. This is essential storage and transmission of digital images. Various image compression techniques have been proposed over the years, and one popular approach is the use of transform. The DWT may be a flag preparing method that breaks down an image its constituent recurrence components. It has been broadly utilized in image compression since it gives a great trade-off between compression proportion image quality [8–11]. Lossless compression: Lossless compressions may be a sort of image compression where all of the information within the image is protected amid the compression prepare. This implies that the image can be superbly reproduced from the compressed information without any misfortune of image quality. Lossless compression procedures ordinarily

accomplish lower compression proportions than lossy compression procedures but are utilized in applications where image quality is basic such as therapeutic imaging and normal images [12–19]. Nanmaran et al. proposed a compression technique and investigated the algorithm with large medical images dataset. The proposed strategy has been assessed on a set of standard test images, but it is critical to test the strategy on a bigger and more differing dataset to decide its adequacy in several scenarios [20]. . In spite of the fact that the proposed strategy has been compared with a few, it is imperative to compare it with other strategies to decide its adequacy completely different scenarios [21]. Expansions to video compression: The proposed strategy can be expanded to compares recordings, and it would be curiously to examine its execution in these applications. Tending to these investigate crevices can lead to the advancement of more productive and viable image compression strategies utilizing Radon transform with discrete wavelet transform [22].

3 Research Methods

Image compression is a critical range of investigate within the field of image handling, because it makes a difference to diminish the measure of image information, making it less demanding to store, transmit and prepare. The Radon transform and discrete wavelet transform (DWT) are two prevalent strategies utilized for image compression [23]. The radon transform is utilized to examine and break down an image into distinctive recurrence groups. One common approach for image compression utilizing Radon transform and DWT is to begin with apply the Radon transform to the image to get projection information. This is known as Radon DWT compression [24]. The Radon transform is the projection of the image intensity along a radial line oriented at a specific angle as shown in Fig. 1. The DWT is utilized to examine and break down the image into diverse recurrence groups. The inverse DWT is the converse operation of the DWT, which changes the flag or image back from the recurrence space to the spatial space [25]. The inverse Radon Transform is taken as input from the set of line integrals obtained from the Radon transform and applies a mathematical operation that reconstruction the original function or image. The process involves back projecting the line integrals onto the image plane using a set of inverse Radon transform filters [26–28].

Image compression could be a process applied to an illustration record to play down its measure in bytes without corrupting image quality underneath a worthy edge. By lessening the record measure, more images can be put away in a given sum of change is utilized to remake an image from its Radon coefficient, which were gotten by anticipating the image onto a set of line integrals utilizing the radon transform. The inverse radon transforms basically turn around the method of the Radon transform and reproduces the initial image from the set of line integrand. The inverse Radon transform can be communicated scientifically as in Eq. 1.

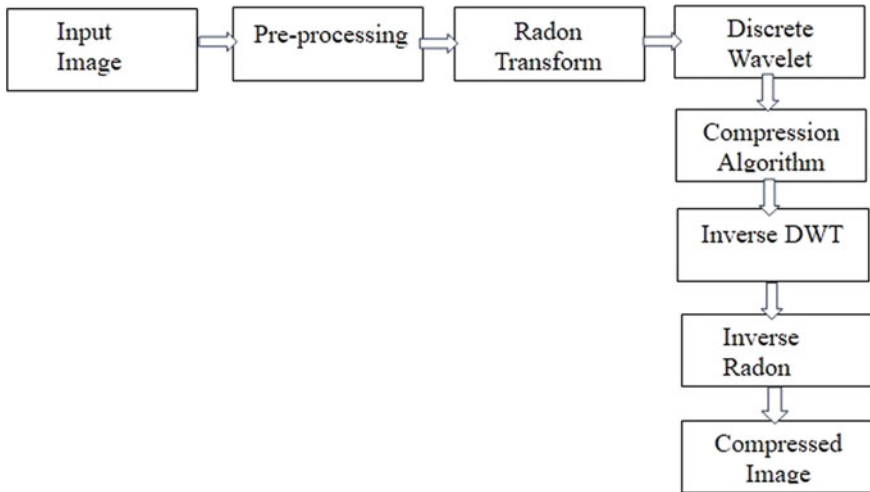


Fig. 1 Proposed methodology (RT + DWT-based image compression technique)

$$f(x, y) = \left(\frac{1}{\pi} \right) \iint R_f(t, \theta) w(x \cos \theta + y \sin(\theta - t)) dt d\theta, \quad (1)$$

where $f(x, y)$ is the reconstructed image, $R_f(t, \theta)$ is the Radon transform of the image, and $w(x)$ is the reconstruction kernel, which is typically chosen to be a high-pass filter such as a ramp filter or a Shepp = Logan Filter [29]. The inverse Radon transform operates by first applying a filter to the Radon transform coefficient, which removes the artifacts caused by the discretization of the line integrals, which is designed to compensate for the blurring effect caused by the projection process. The inverse Radon transform is a mathematical technique used in image compression to reconstruct an image from its Radon transform coefficients. It reverses the process of the Radon transform and back projects the filtered coefficients onto the image plane using a reconstruction image [30].

3.1 Radon Transform Method (RT)

The Radon transform may be a numerical method that based on a function f and another function R_f which defined on a two dimensional space . In image compression, the Radon transform is regularly utilized as a pre-processing step to extricate highlights from an image that can be utilized to speak to the image in a compressed form. The Radon transform can be expressed mathematically as in Eq. 2 [31–33].

$$R_f(t, \theta) = \iint f(x, y) \delta(x \cos \theta + y \sin(\theta - t)) dx dy, \quad (2)$$

where $f(x, y)$ is the original image, $R_f(t, \theta)$ is the Radon transform of the image at a distance t and angle θ , and δ is the Dirac delta function. The Radon transform essentially maps the image function onto a domain that consists of the set of line integrals [34–36].

4 Results and Discussion

The simulation results obtained from the proposed technique are discussed and compared with existing techniques in this section.

4.1 Simulation Results of Proposed Hybrid Compression Technique

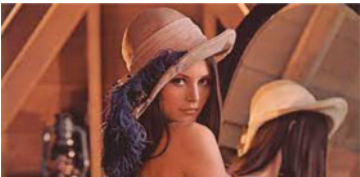
For test investigation, nine images were taken from the image database in arrange to confirm the productivity of proposed image compression strategy Radon transform (RT) with discrete wavelet transform (DWT). The image insights to be specific SSIM, CR, and PSNR are utilized to analyze the first image taken for this assessment uncovered in Fig. 2.

The image compression technique that combines Radon transform with discrete wavelet transform (DWT), as shown in Fig. 2, has been applied to reconstruct the image. The reconstructed image has been evaluated using various performance metrics, such as PSNR, SSIM, CR, and the results are presented in Table 1. The reconstructed image has achieved a PSNR value of 28.73 dB, CR value of 88.33%, and an SSIM value of 89.41%.

Fig. 2 Sample original image (vegetables)



Table 1 Performance measures of image compression using Radon transform (RT) with discrete wavelet transform (DWT) technique

S. No.	Sample input image	CR (%)	PSNR (dB)	SSIM (%)
1	 Vegetables	88.33	28.73	89.41
2	 Lena	83.33	25.48	86.03
3	 Penguins	80.45	25.43	84.06
4	 Camera man	78.35	25.03	83.11
5	 Hlevkin	76.37	24.55	83.78
6	 Butterfly	75.55	23.76	82.87

The Radon transform is mathematical operation that maps a line in an image to a point in the Radon transform space, essentially converting the line singularity to a point singularity. This mapping is done using the Radon transform Eqs. (2) and (1) as part of the image compression algorithm in a gray scale image, as shown in Fig. 3. The Radon points obtained from this mapping, which represent the line features in the image, are shown in Fig. 4. These Radon points are generated using the Radon function, and they capture the information about the lines present image.

The Radon transform is a mathematical operation that maps lines in an image to points in a Radon space. The resulting Radon transform points are then encoded utilizing discrete wavelet transform (DWT). At the receiver side, the encoded Radon points are decoded using inverse DWT. The combination of Radon transform and DWT forms a compression algorithm that produces two levels of outputs, as shown in Figs. 5 and 6. Wavelet decomposition level 1 is divided into four parts A1, H1, V1, D1, and wavelet decomposition level 2 is divided into six parts.

The reconstructed image is obtained by applying the inverse discrete wavelet transform (IDWT) to the encoded Radon points, as shown in Figs. 5 and 6. The performance of the proposed image compression scheme is evaluated using several metrics, including PSNR, SSIM, and CR. The PSNR, SSIM, and CR value of proposed image compression technique using Radon transform with DWT is 28.73 dB, 89.41%, and

Fig. 3 Gray scale image



Fig. 4 Radon transform on input image



Fig. 5 Wavelet decomposition level 1

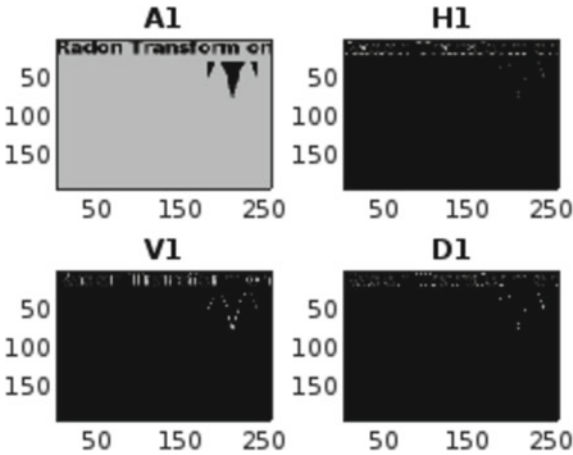
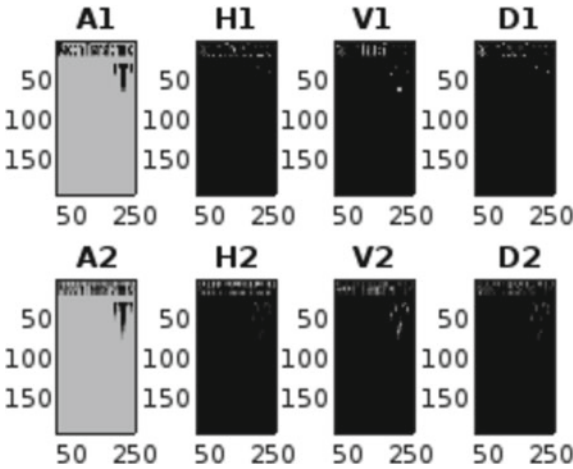


Fig. 6 Wavelet decomposition level 2



83.33%. Further, the analysis of proposed technique has tabulated below (Fig. 7; Tables 2, 3 and 4).

Fig. 7 Compressed image



Table 2 PSNR, SSIM, and CR values for varying Radon transform angle on vegetable image

S. No.	Radon angle	CR (%)	PSNR (dB)	SSIM (%)
1	30	88.33	28.73	89.41
2	60	88.45	29.21	90.07
3	90	88.97	29.89	90.98
4	120	89.34	30.08	91.58
5	150	89.78	30.98	92.87
6	180	90.01	31.68	93.78
7	210	90.34	32.98	94.49
8	240	90.89	33.78	95.88
9	270	91.34	34.67	96.74

Table 3 Comparison between Proposed Hybrid (RT + DWT) approach with individual techniques

Performance metrics	Discrete wavelet transform (DWT)	Radon transform (RT)	DWT + Radon transform (proposed work)
Compression ratio (%)	83.33	48.014	88.39
Peak signal-to-noise ratio (dB)	27.56	36.898	45.41
Structural similarity index (%)	85.18	71.692	89.03

Table 4 Comparison between Proposed Hybrid (RT + DWT) approach with existing compression techniques

Compression techniques	PSNR	SSIM	CR
Discrete cosine transform [37]	37.82	81.3	–
Discrete wavelet transform [38]	34.70	–	27.54
Gabor transform [39]	44.64	45	–
Discrete shearlet transform [40]	39.59	1	–
DWT + Radon transform (proposed work)	45.41	89.03	88.39

5 Conclusion

Picture compression utilizing Radon transform with discrete wavelet transform is a successful technique for reducing the storage and transmission requirements of standard images. By applying the Radon transform to an image to obtain its sinogram, and then applying discrete wavelet transform to the sinogram, the image can be effectively compressed while preserving its essential features. The compression algorithm can be further optimized by adjusting the parameters of the compression algorithm to achieve a balance between Compression Ratio and image quality that is suitable for a particular application. Performance measures such as PSNR, SSIM, and CR can be used to evaluate the effectiveness of the compression algorithm. Such as like Vegetables values $CR (\%) = 83.33$, $PSNR (dB) = 28.73$, and $SSIM (\%) = 89.41$. In the future there is scope for further research into the development of new image compression algorithm that combine Radon transform with other techniques such as deep learning and neural networks. Additionally, there is scope for research into the development of new performance measures that can more accurately evaluate the effectiveness of image compression algorithm in real-world applications. Picture compression utilizing Radon transform with discrete wavelet transform may be an effective strategy that has numerous potential applications in areas such as a medication and inaccessible detecting.

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